

Functional requirements:

user should be able to solve any problem based on his preference,
user should be able to search for any problem ,
user should be able to participate in coding contest based on his preference,

Non-functional requirements:

Data-partitioing,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here We need Strong Consistency for rank predictions during contests ,
& for search scenario we need to prioritise 99.99% uptime i.e Availability>>Consistency)

Core Entities:

user ,
problem,
contest,
solution

API :

POST:/submit/solution

{

 userID,

 {FileMetadata}

}

GET:/search/{problemID}

HLD :

